DEFENSE LOGISTICS AGENCY
LAND AND MARITIME
P.O. BOX 3990
COLUMBUS, OHIO 43218-3990

DLA LAND AND MARITIME-VAI 18 August 2025

MEMORANDUM FOR MILITARY/INDUSTRY DISTRIBUTION

SUBJECT: Initial Drafts of various MIL-DTL-38999 Slash Sheets
(see table below for details).

These initial drafts for these subject documents are now available for viewing and downloading from the DLA Land and Maritime-VA Web site:

https://landandmaritimeapps.dla.mil/Programs/MilSpec/DocSearch.aspx

Initial Draft	Title	Project Number
MIL-DTL-38999/10C	Connectors, Electrical, Circular, Receptacle, Dummy Stowage, Bayonet Coupling, (MIL-DTL-27599, Series II and MIL-DTL-38999, Series II)	5935-2025-086
MIL-DTL-38999/20J	Connectors, Electrical, Circular, Receptacle, Threaded, Wall Mounting Flange, Removable Crimp Contacts, Series III, Metric	5935-2025-087
MIL-DTL-38999/21D	Connectors, Electrical, Circular, Threaded, Receptacle, Box Mounting Flange, Hermetic, Hermetic Solder Contacts, Series III, Metric	5935-2025-088
MIL-DTL-38999/22K	Connectors, Electrical, Circular, Receptacle, Threaded, Dummy Stowage, Series III, Metric	5935-2025-089
MIL-DTL-38999/23D	Connectors, Electrical, Circular, Threaded, Receptacle, Jam-Nut Mounting, Hermetic, Hermetic Solder Contacts, Series III, Metric	5935-2025-092
MIL-DTL-38999/24K	Connectors, Electrical, Circular, Receptacle, Threaded, Jam-Nut Mounting, Removable Crimp Contacts, Series III, Metric	5935-2025-093
MIL-DTL-38999/25D	Connector, Electrical, Circular, Threaded, Receptacle, Solder Mounting, Hermetic, Solder Contacts, Series III, Metric	5935-2025-094
MIL-DTL-38999/26H	Connectors, Electrical, Plug, Circular, Threaded, Straight, Removable Crimp Contacts, Series III, Metric	5935-2025-095
MIL-DTL-38999/27D	Connectors, Electrical, Circular, Threaded, Receptacle, Weld Mounting, Hermetic, Hermetic Solder Contacts, Series III, Metric	5935-2025-096
MIL-DTL-38999/28J	Connectors, Electrical, Circular, Nut, Hexagon, Connector Mounting, Series III and IV, Metric	5935-2025-097
MIL-DTL-38999/29D	Connectors, Electrical, Circular, Threaded, Plug, Lanyard Release, Fail-Safe, Removable Crimp Contacts, Pins, Series III, Metric	5935-2025-098
MIL-DTL-38999/30D	Connectors, Electrical, Circular, Threaded, Plug, Lanyard Release, Fail-Safe, Removable Crimp Contacts, Sockets, Series III, Metric	5935-2025-099

Major changes to these documents include removing dimensional equivalents.

Concurrence or comments are required at this Center within 30 days. Late comments will be held for the next coordination of the document. Comments from military departments must be identified as either "Essential" or "Suggested". Essential comments must be justified with supporting data. Military review activities should forward comments to their custodians of this office, as applicable, in sufficient time to allow for consolidating the department reply. Lack of response to this draft will be construed as concurrence.

If these documents are of interest to you, please provide your comments or suggested changes. The point of contact for this document is Mr. Bender, phone number 614-692-9185, facsimile transmission, 614-692-6939, e-mail Jacob.Bender@dla.mil or may be mailed via the US Postal Service to DLA LAND AND MARITIME, ATTN: VAI (Attention: Jacob Bender), P.O. Box 3990, Columbus, OH 43218-3990.

Sincerely,

/ SIGNED /

ERIC CANDELARIA
Acting Chief,
Interconnection Branch

CC: David Devine, Robert Schoening, Alan Drais, Eddie Shen, Deborah Thompson

Note: This initial draft, dated August 22, 2025, prepared by DLA-CC has not been approved and is subject to modification. DO NOT USE FOR ACQUISITION PURPOSES. (Project: 5935-2025-098)

METRIC

MIL-DTL-38999/29D
DRAFT
SUPERSEDING
MIL-DTL-38999/29C
w/AMENDMENT 2
11 August 2022

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, PLUG, LANYARD RELEASE, FAIL-SAFE, REMOVABLE CRIMP CONTACTS, PINS, SERIES III, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.

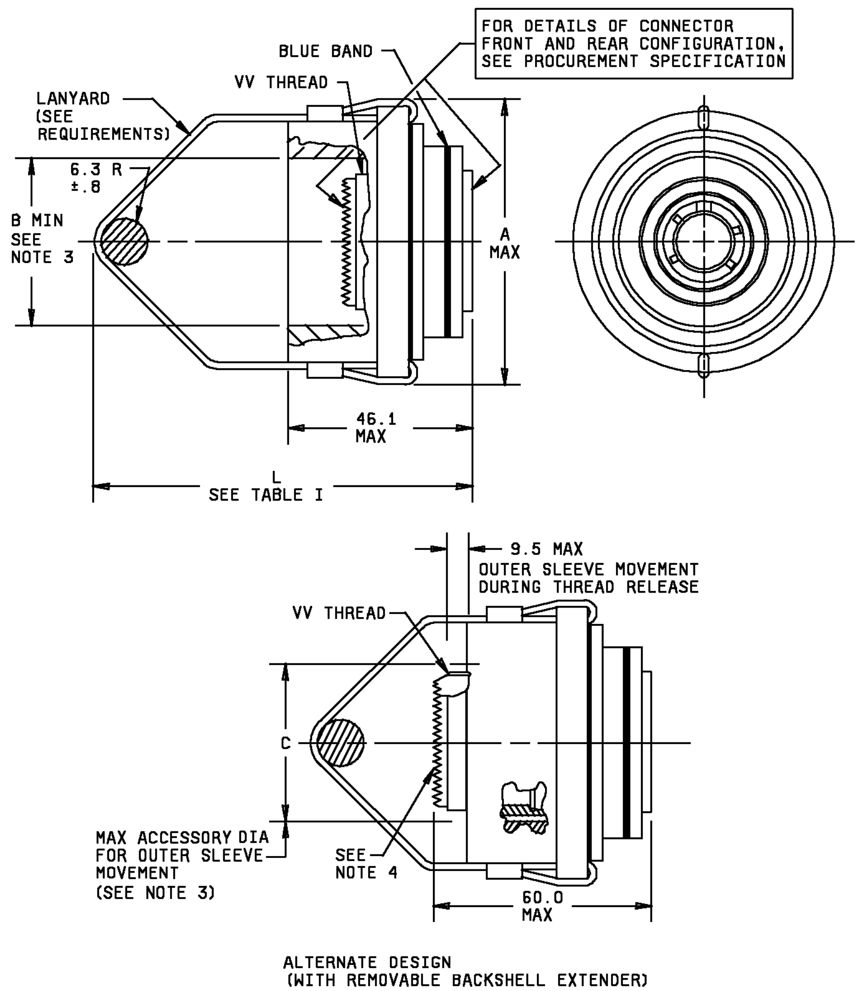

FIGURE 1. Plug.

AMSC N/A

FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Shell size	Shell size code	A max	B min	C max	VV thread	Backshell extender torque
11	B	46.9	25.5	25.0	M15x1.0-6g 0.100R	5.6 Newton-meters min
13	C	50.1	29.9	29.4	M18x1.0-6g 0.100R	
15	D	52.8	33.0	32.5	M22x1.0-6g 0.100R	
17	E	56.0	36.2	35.7	M25x1.0-6g 0.100R	
19	F	59.2	39.0	38.5	M28x1.0-6g 0.100R	
21	G	62.8	42.2	41.7	M31x1.0-6g 0.100R	
23	H	65.9	45.4	44.9	M34x1.0-6g 0.100R	
25	J	68.7	48.5	48.0	M37x1.0-6g 0.100R	

NOTES:
1. Dimensions are in millimeters.
2. EMI grounding feature required on this connector.
3. Dimension indicates clearance required for proper operation. Connector must be capable of accepting and functioning with applicable SAE-AS85049 accessories.
4. Backshell extender is factory installed but may be removed for serviceability if necessary. If removed the extender must be reassembled and torqued to specified value for proper operation of connector.

FIGURE 1. Plug - Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1. Interface dimensions shall conform to MIL-DTL-38999.

This connector mates with MIL-DTL-38999/20, /21, /22, /23, /24, /25 and /27.

For insert arrangements: See MIL-STD-1560.

Lanyard:

a. 1.57 millimeter (.06 inches) minimum diameter, seven strands of stainless steel capable of withstanding 890 Newton pull test after assembly with connector.
b. Cable shall be covered with a suitable protective sleeving to preclude possible chafing or abrading of wires.

Connector shall disengage from any coupling condition including partially mated.

Connector design shall incorporate a swivel action for the lanyard to prevent twisting of the cable.

Part or Identifying Number (PIN) example:

D38999/ 29 W B 35 E N

D38999/	DoD number prefix
29	Specification sheet number

W Class
B Shell size code
35 Insert arrangement
E Lanyard length code (see table I)
N Polarizing positions (N is required for normal position)

Class F is not for Navy use and is inactive for Air Force new design use.

Note: The term PIN is equivalent to the term (part number, identification number, and type designator) which was previously used in this specification.

TABLE I. Lanyard length codes.

Code	L ± 6 mm (.24)	Code	L ± 6 mm (.24)
A	102 (4)	M	254 (10)
B	115 (4.5)	N	267 (10.5)
C	127 (5)	P	280 (11)
D	140 (5.5)	R	293 (11.5)
E	153 (6)	S	305 (12)
F	166 (6.5)	T	318 (12.5)
G	178 (7)	U	331 (13)
H	191 (7.5)	V	356 (14)
I	203 (8)	W	381 (15)
J	216 (8.5)	X	407 (16)
K	229 (9)	Y	432 (17)
L	242 (9.5)	Z	458 (18)

Qualification: Connectors shall meet the qualification requirements of MIL-DTL-38999 with the exceptions and additions specified below. Group C periodic requalification testing, including the selection of Group C test samples, shall be in accordance with testing specified in this specification sheet and MIL-DTL-38999 Group C, 24 month testing.

Durability:

Wired connectors shall meet the durability requirements of MIL-DTL-38999, with the following exception: Total number of cycles of normal mating and unmating shall be 250 (200 cycles of normal mating and unmating followed by 50 cycles of normal mating with pull-separation unmating).

Pull-separation at temperature:

In addition to mating and unmating by normal coupling ring rotation, the connector shall be capable of lanyard-pull separation at any angle within 15° of the normal axis after exposure for 1 hour minimum at the following temperatures as shown in table II: Room ambient temperature, -65°C (+ 0°C / -5 °C), and at the maximum temperature of the specified class. Each connector shall have one straight (0°) pull and one pull at 15° from straight, with a pull rate not exceeding 13 cm/second. Each pull test shall be performed within three (3) minutes of removal from the temperature chamber without forced heating or cooling. Maximum separation forces shall be as specified in table III.

TABLE II. Separation test temperatures.

Pull Type	Step 1	Step 2	Step 3
Straight, 0° Pull	Room ambient	-65° + 0° / -5 °C	max temperature of specified connector class
15° Pull	Room ambient	-65° + 0° / -5 ° C	max temperature of specified connector class

TABLE III. Separation forces (max).

Shell size	Straight pull (N)	15° pull (N)
11	200	245
13	200	245
15	200	245
17	400	445
19	400	445
21	400	445
23	400	445
25	400	445

Fail-safe disengagement:

Connectors shall be partially mated with the plug coupling ring rotated approximately 50% of full coupling. Pull-separation both straight and 15° from straight at ambient temperature shall be accomplished within the limits specified in table III.

Vibration:

Wired mated connectors shall meet the vibration requirements of MIL-DTL-38999 with the following exceptions:

a. Sine vibration: Connectors shall be subjected to the test specified in MIL-STD-202, method 204, test condition G. Accessory load shall be omitted.
b. Random vibration: Connectors shall be subjected to the test specified in EIA-364-28, test condition VI, letter “J”, ambient temperature. Duration shall be 8 hours in the longitudinal direction and 8 hours in a perpendicular direction, for a total of 16 hours. Accessory load shall be omitted. Additional random vibration tests as specified in MIL-DTL-38999 shall be performed.

Ice resistance:

The mated, wired connectors with accessories attached shall be placed in a chamber and the temperature reduced and stabilized such that the item is maintained at -18°C (tolerance of +0°C, -5°C) for 1 hour. After stabilization of the chamber temperature, the test item shall be sprayed with water precooled to 2°C (tolerance of +5°C, -0°C), for a period of five (5) minutes. The test item shall be located a maximum of 305 millimeters (12 inches) from the spray nozzle. The entire test item shall be exposed to the spray. After completion of water spray, the test item shall remain in the chamber at -18°C (tolerance of +0°C, -5°C), for an additional 30 minutes. Upon completion of the 30 minute cold soak period, the test item shall be removed from the chamber and immediately (within two (2) minutes) subjected to uncoupling by use of the lanyard mechanism. The force required to separate the connector shall not exceed the values in table II by more than 50%.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-DTL-38999/20
MIL-DTL-38999/21
MIL-DTL-38999/22
MIL-DTL-38999/23
MIL-DTL-38999/24
MIL-DTL-38999/25
MIL-DTL-38999/27
MIL-STD-202
MIL-STD-1560
EIA-364-28
SAE-AS85049

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy - AS
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2025-098)

Review activities:
Army - AR, MI
Navy - EC, MC, OS
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.